

Performance Testing

Date	30 June 2026
Team ID	
Project Name	Credit Card Approval Prediction Using Machine Learning
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Flask Prediction API (/predict)
Test Environment	Hosted Server (PythonAnywhere)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load Home Page	1	30	Home page loads successfully
2	Submit Valid Applicant Details	5	60	Prediction is displayed correctly
3	Multiple Prediction Requests	10	60	Server handles concurrent requests without errors

4	Invalid Input Testing	5	30	Validation message is displayed without crashing
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	Fast response time
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Within acceptable limit
3	Throughput (Req/sec)	5 Req/sec	7 Req/sec	Pass	Stable under normal load
4	Error Rate	< 1%	0%	Pass	No request failures
5	CPU Utilization	< 80%	45%	Pass	Normal server usage
6	Memory Utilization	< 80%	52%	Pass	Memory usage within limit

Step 4: Observations & Analysis

- ☑ The application responded quickly under normal user load.
- ☑ All prediction requests were processed successfully.
- ☑ No server crashes or unexpected errors were observed during testing.
- ☑ CPU and memory usage remained within acceptable limits.
- ☑ The Flask application deployed on PythonAnywhere handled multiple requests efficiently and produced consistent prediction results.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

